

- 1 A Si p⁺-n-p transistor has impurity concentrations of 5×10^{18} , 2×10^{17} and 10^{16} cm⁻³ in the emitter, base, and collector, respectively. The base width W_B is 1 μ m and the device cross-sectional area is 2 mm². When the emitter-base junction is forward-biased to 0.5 V and the base-collector junction is reverse-biased to 5 V, calculate (a) the neutral base width and (b) the minority carrier concentration at $x=0$ in the neutral base.

(Ans. 0.904 μ m and 1.13×10^{11} cm⁻³)

- 2 For the transistor in Problem 1, the diffusion constants of minority carriers in the emitter, base, and collector are 52, 40, and 115 cm²/s, respectively; and the corresponding lifetimes are 10^{-8} , 10^{-7} , and 10^{-6} s. Find the current component I_{Ep} , I_{Cp} , I_{En} , I_{Cn} , and I_{BB} .

(Ans. 1.596×10^{-5} A, 1.596×10^{-5} A, 1.04×10^{-7} A, 3.196×10^{-14} A, 0)

- 3 Using the results obtained from problem 1 and 2, (a) find the terminal currents I_E , I_C , and I_B , of the transistor; (b) calculate emitter efficiency, base transport factor, common-base current gain, and common-emitter current gain. (c) Comment on how the emitter efficiency and base transport can be improved.

(Ans. (a) 1.606×10^{-5} A, 1.596×10^{-5} A, 1.041×10^{-7} A, (b) 0.9938, 1, 0.9938, 160.3)

4. (a) A Si transistor has D_p of 10 cm² and W of 0.5 μ m. Find the cutoff frequencies for the transistor with a common base current gain of 0.998. Neglect the emitter and collector delays.
- (b) To design a bipolar transistor with 5 GHz cutoff frequency. What the neutral base width W will be?

(Ans: (a) $f_T = 1.27$ GHz and $f_\beta = 2.55$ MHz; (b) 0.252 μ m)